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Do we all feel the city the same way? A proof-of-concept framework linking physiological wellbeing to neighborhood-scale resilience through multi-domain environmental sensing

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ABSTRACT

Outdoor comfort and climate adaptation studies usually focus on thermal-only investigations, typically relying on straightforward heat stress metrics. Human-centered validation frameworks to confirm whether built environment interventions reduce actual physiological stress versus predicted discomfort from population-averaged models still lack. To fill this gap, this proof-of-concept addresses three questions: (1) Do multi-domain environmental models explain physiological variance that thermal indices alone cannot capture? (2) Do individuals exhibit heterogeneous resilience to identical landscapes? (3) How can individual physiological data aggregate into neighborhood-scale resilience zones? Using synchronized wearable sensors (heart rate, electrodermal activity) and multi-domain environmental monitoring (Universal Thermal Climate Index, Particulate Matter, CO_2, Acoustic Spectrum), we tested eight participants through contrasting Roman urban typologies such as exposed piazzas, tree-canopied riverbanks, enclosed gardens. Thermal comfort indices alone failed to predict stress, yet multi-domain models integrating thermal, air quality, and acoustic parameters explained up to 43% of physiological variance. Individuals with similar characteristics (age 28–38) and lifestyle exhibited markedly different physiological responses to identical microclimates, suggesting that the city is not felt the same way by everyone, and that population-averaged vulnerability models may not reliably predict individual-level stress outcomes. We operationalized three resilience metrics: Toughness, Recovery Capacity, Adaptation Duration, derived through generalized logistic growth modeling capturing nonlinear threshold dynamics in physiological stress responses. K-means clustering identified five statistically distinct adaptation zones corresponding to landscape typologies, demonstrating that individual wearable data enables neighborhood-scale resilience mapping. This framework establishes a preliminary methodological infrastructure for integrating physiological sensing with urban digital twins, enabling validation of whether landscape configurations reduce stress for inhabitants rather than relying on demographic prediction alone.

1. Introduction

Urban areas house 55% of the world's population, projected to reach 68% by 2050 [1], intensifying challenges in designing built environments that maintain thermal comfort, air quality, and acoustic performance under changing climatic conditions. While initiatives like C40

Abbreviation: UTCI, Universal Thermal Climate Index; CCA, Climate Change Adaptation; T, Toughness; AC, Ability to Cope; RC, Recovery Capacity; PM, Particulate Matter; BPM, Beats per Minute; EDA, Electrodermal Activity; UAQI, Universal Air Quality Index; UHI, Urban Heat Island Effect; LST, Land Surface Temperature; ART, Attention Restoration Theory; SRT, Stress Recovery Theory; GIS, Geographic Information System; IoT, Internet of Things; LOPO, Leave-One-Participant-Out; CV, Coefficient of Variation; RMS, Root Mean Square.

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Cities [2] and the EU Covenant of Mayors [3] emphasize Climate Change Adaptation (CCA) planning, current building science approaches lack methods to validate whether environmental interventions actually reduce physiological stress for individual occupants versus predicted discomfort from population-averaged models.

Simplified threshold-based approaches, while scalable, cannot capture real-time physiological adaptation dynamics determining actual health risk [4]. Emerging evidence shows that compound environmental stressors such as heat combined with air pollution and noise can produce interaction effects unpredictable from single-stressor models, necessitating integrated multi-domain approaches [5,6]. Structural equation modeling reveals that audio-visual interactions ($\beta = 0.57$) of environmental satisfaction, with thermal, acoustic, and visual domains operating through bidirectional sensory influences rather than independent channels [7]. Yet conventional building design treats thermal conditioning, ventilation, and acoustic control as independent systems rather than synergistic determinants of occupant wellbeing. The resulting built environments may satisfy individual performance standards while failing to support actual physiological resilience, a contradiction undetectable without measuring inhabitant stress responses. Multi-domain resilience operates through parallel analytical scales and physiological regulatory systems responding to different environmental stressors, which interact synergistically rather than additively [8,9].

Recent advances in wearable sensor integration with urban IoT networks enable continuous, synchronized collection of individual physiological data alongside hyperlocal environmental conditions, providing unprecedented resolution for linking personal stress responses to specific urban microclimates [10]. Wearable technologies measuring heart rate and electrodermal activity (EDA) [11] transform climate resilience assessment by enabling real-time validation of design interventions. In a recent study, researchers used EDA wristbands to track students' skin conductance during math tasks under varying classroom microclimates (temperature, CO2, PM). Higher CO2 and temperatures correlated with increased EDA features like Conductance Level (CL), indicating stress; machine learning confirmed temperature and air quality's role in physiological well-being and thermal comfort [12]. However, population-targeted urban sensing approaches perpetuate the limitation of treating vulnerability as spatially uniform rather than recognizing heterogeneous exposure risks across population subgroups [13,14]. Landscape spatial patterns within 175-meter radius buffers exert greater influence on soundscape perception than on-site composition [15]. While urban morphology represents relatively fixed constraints amenable to Geographical Information System (GIS)-based thermal comfort modeling and soundscape mapping [16,17], human physiological adaptation remains highly flexible and individualized, necessitating integration of population-scale landscape interventions with person-scale physiological sensing. The present study advances beyond existing individual-level outdoor comfort research in three critical dimensions. First, while recent walking experiments have successfully employed wearable sensors to measure physiological responses to environmental sequences and demonstrated the restorative effects of green exposure [18], these studies typically focus on route-level or scenario-level comparisons rather than quantifying individual-specific resilience profiles across multiple environmental domains. Second, although emerging frameworks integrate multi-domain urban features including visual, acoustic, and microclimatic variables to predict neighborhood-scale comfort indices using explainable AI [19]), these approaches aggregate population-level perceptual data rather than measuring individual physiological stress responses that determine actual health risk. Third, smartwatch-based interventions have demonstrated behavioral nudging potential for thermal and acoustic comfort [20], yet these implementations assess perceived usefulness and behavior change rather than validating whether interventions actually reduce physiological stress burden as measured through dynamic recovery or cumulative allostatic markers. No framework currently integrates continuous multi-domain environmental sensing with individual-level physiological monitoring to validate whether interventions reduce measured stress for actual inhabitants. Our methodological contribution lies in integrating continuous multi-domain environmental sensing (thermal, acoustic, air quality) with individual-level physiological monitoring to: (1) validate whether multi-domain models predict acute stress responses significantly better than single-domain thermal comfort indices, (2) quantify heterogeneous resilience profiles among physically similar individuals, examining whether person-specific physiological response patterns exist even after controlling for environmental variation, and (3) develop a framework for aggregating individual sensing data into neighborhood-scale resilience zones that can inform spatially-targeted built environment interventions.

This proof-of-concept study addresses a critical gap: the absence of individualized physiological data in validating outdoor environmental performance of built environments. Resilience operates across temporal scales. For example, dynamic recovery [21] quantifies acute physiological restoration rates post-exposure, allostatic load (long--term-exposure [22]) captures cumulative regulatory system dysregulation, and adaptation time measures thermoregulatory stabilization periods. While vulnerability assessments typically focus on extreme thermal conditions, we examine resilience under non-extreme exposure where multisensory interventions can support physiological restoration [23]. Three research questions guide this investigation. First, do multi-domain environmental models integrating thermal, acoustic, and air quality parameters predict individual physiological stress responses significantly better than single-domain thermal comfort indices? This question addresses whether the integrated sensing approach provides superior explanatory power compared to conventional thermal-only assessment, validating the multi-domain framework's added value. Second, do individuals exhibit heterogeneous resilience profiles to identical landscape conditions, challenging the validity of demographic vulnerability assumptions in building performance assessment? This examines whether person-specific physiological response patterns exist even after controlling for environmental variation, establishing the necessity of individualized rather than population-averaged approaches. Third, how can neighborhood-scale resilience zones be identified through aggregated individual sensing data? This bridges from individual-level heterogeneity to spatial planning applications, demonstrating how personalized physiological monitoring can inform area-based intervention strategies. Using synchronized wearable physiological sensors and multi-domain environmental monitoring, we developed a framework integrating individual-level resilience metrics that can be aggregated to neighborhood scale, demonstrating how these insights inform human-centered urban digital twins and evidence-based multisensory design strategies [24] for the built environment [25].

2. Methodology

This proof-of-concept framework examines individual physiological resilience to multi-domain environmental stressors through synchronized wearable sensing and hyperlocal monitoring during structured urban transect walks. Data collected during these walks were analyzed through four stages (Fig. 1): (1) correlation analysis quantifying relationships between physiological indicators (heart rate, EDA) and environmental parameters (thermal, acoustic, air quality domains) to establish baseline associations and inter-individual variability; (2) generalized logistic growth modeling fitted to physiological time-series under single-domain and multi-domain (thermal + acoustic + air quality) configurations to compare explanatory power; (3) transformation of model parameters into three resilience metrics: Toughness (resistance to stressors), Recovery Capacity (speed of baseline return), and Adaptation Duration (stabilization period), operationalizing Schweiker's resilience framework [26]; and (4) K-means clustering on temporally-weighted, recovery-coupled environmental features to

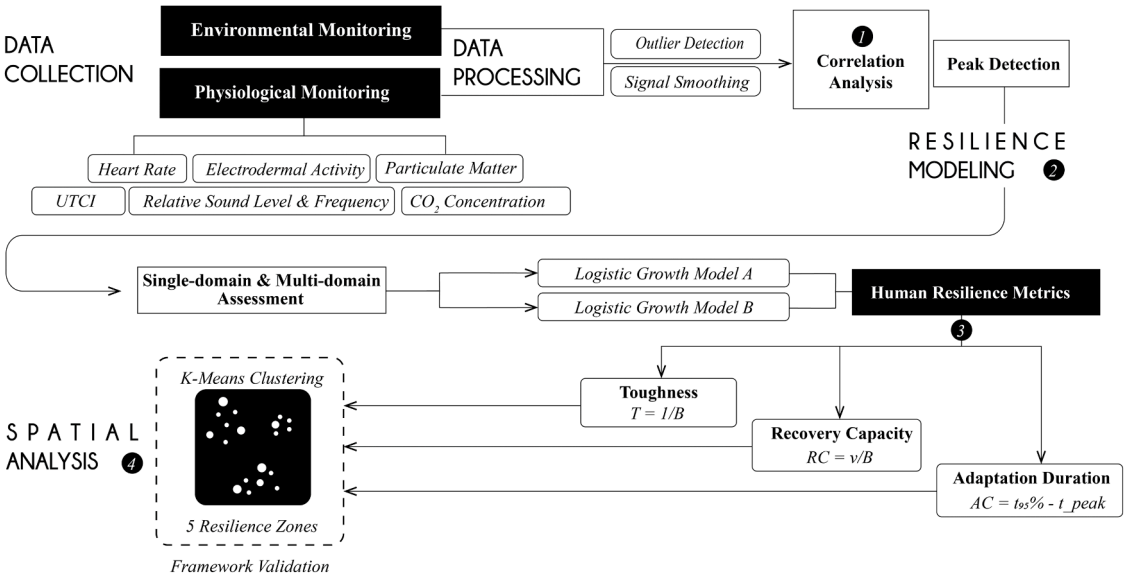


Fig. 1. Methodology.

identify spatially distinct adaptation zones. This framework enables direct comparison of thermal-only versus integrated environmental models, characterizes individual-specific adaptation profiles, and demonstrates aggregation of personalized physiological data into neighborhood-scale resilience maps corresponding to measurable urban typologies.

2.1. Data collection

2.1.1. Experimental design

Data collection took place through a multi-domain comfort walk, intentionally conducted during non-heatwave conditions, as existing literature suggests that extreme heat often mask subtle multi-domain effects and interactions between environmental factors, making

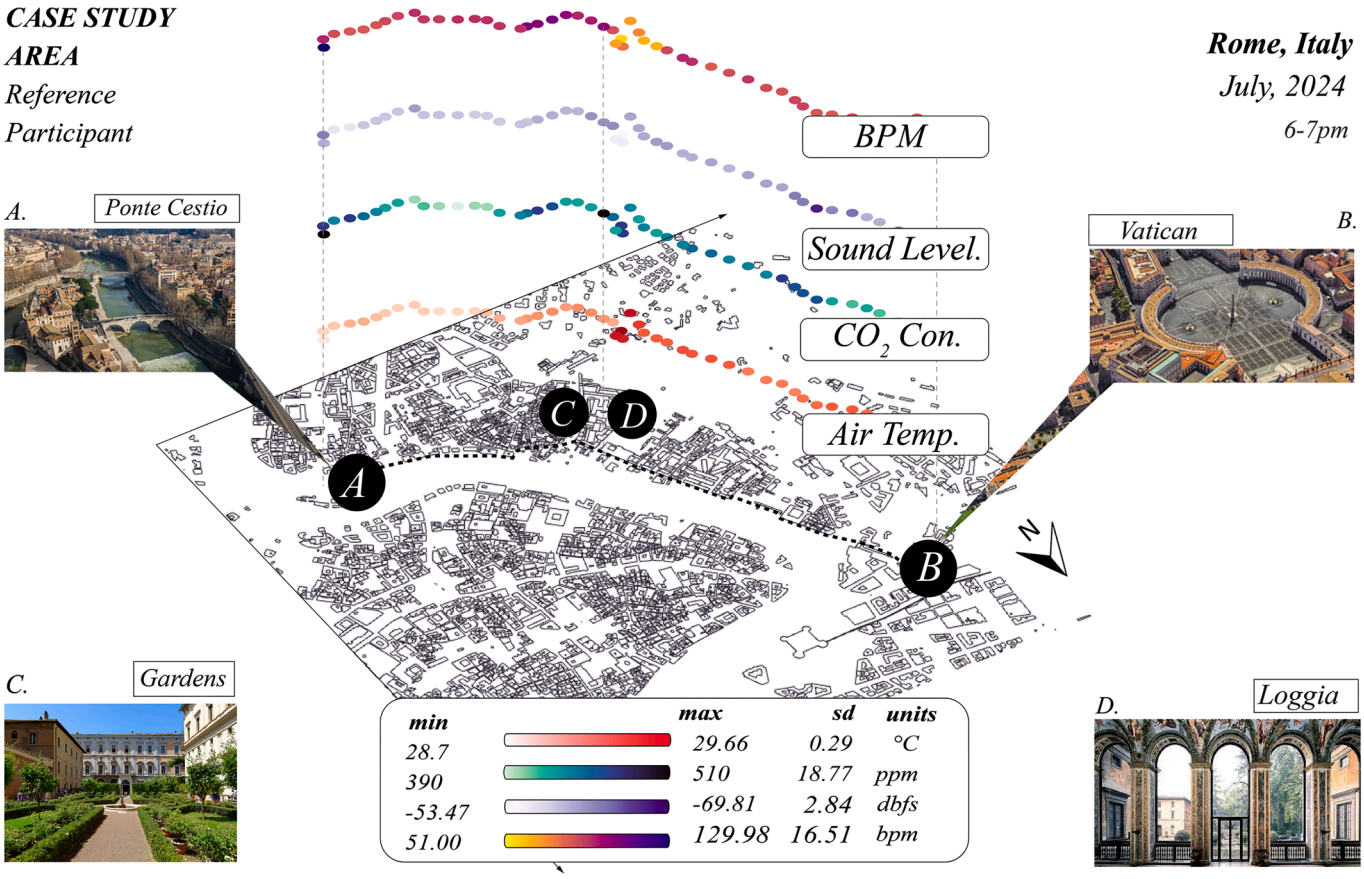


Fig. 2. Case study area and overview of the multi-domain comfort walk and key variables per domain retrieved during the experimental campaign.

humans less sensitive to non-thermal stimuli such as acoustics or visual [27–30]. During each session of the comfort walks, two researchers carried wearable environmental monitoring backpacks [31], while two participants wore Empatica medical devices to measure heart rate, EDA, and skin temperature [32]. Each session followed a structured route (Fig. 2), where one participant and one researcher commenced their walk from an area characterized by elevated thermal discomfort and higher pollution levels, progressing toward an environment with more favourable environmental conditions, while the other participant and researcher walked in the opposite direction. The initial selection of transect start and end points (A and B in Fig. 2) was informed by Land Surface Temperature (LST) data derived from Landsat 8 and the Universal Air Quality Index (UAQI) obtained from the Google Air Quality API. This methodological design facilitated the examination of spatio-temporal variations in environmental exposure and human perception across distinct urban microclimates.

The one-hour lasting comfort walk was performed in Rome, Italy, with a hot-summer Mediterranean climate (Csa) according to the Köppen climate classification [33] between July 1st and 2nd of 2024. The " comfort walk" protocol [34,35] instructed participants to maintain their natural, self-selected walking pace typical of urban pedestrian movement, without running or prolonged standing. All participants completed the one-hour route covering approximately 2.5 km under researcher supervision to ensure comparable pacing. The starting and ending points of the transect are represented in Fig. 2. Point B, located in St. Peter's Square, Vatican City, is an open, highly exposed environment with minimal vegetation, with LST = 43.9 °C and UAQI=135.37, while Point A, situated at Ponte Cestio across the Tiber River, features a denser and taller tree canopy, offering greater thermal buffering with LST=37.9 °C and UAQI=126.40 on average during the previous week before the experimental campaign. In the midpoint of the route participants crossed a garden fenced with brick walls of 4 m and dense vegetation (Fig. 2, Point C) and a 'transitional space' (Fig. 2, Point D) which in the case of this experiment was a *'loggia: a room or gallery with one or more open sides, especially one that forms part of a house and has one side open to the garden'* [36] to account for similar architectural typologies that are often present in the case study area.

The study participants comprised university students and academic personnel aged 28–38 years, representing diverse international backgrounds. This sample was selected to ensure a relatively homogeneous sample, excluding individuals from recognized vulnerable populations [37]. The narrow age range (28–38) and similar educational/occupational backgrounds minimize confounding from major demographic factors, though residual variability from unmeasured covariates (sleep quality, caffeine intake, recent stress exposure, menstrual cycle phase) remains a limitation of within-subject models.

Each session followed four phases. First, pre-walk preparation (~15 min) included informed consent, wristband equipment, and walking instructions. Second, stationary assessment periods (5–10 min) occurred at Points A, B, C, D where participants paused within 5 m of the environmental backpack, acclimatized for 15 min, then completed questionnaires via their smartphones. Third, walk segments consisted of natural-pace walking with the researcher carrying the environmental backpack maintaining $\leq$2 m proximity and continuous monitoring. Fourth, session completion involved final questionnaires. Bidirectional design enabled one pair to walk A→C→D→B while another walked B→C→D→A simultaneously. The two-day experimental period was designed as a proof-of-concept validation study to test whether multi-domain models predict physiological responses better than thermal indices alone, and whether physically similar individuals exhibit heterogeneous resilience profiles. The short timeframe enabled controlled comparison across contrasting urban typologies under stable meteorological conditions, minimizing confounding variables for initial framework validation.

2.1.2. Environmental measures

The wearable monitoring system was designed and developed at the University of Perugia as a backpack to ensure ergonomic comfort while allowing comprehensive environmental data collection on a hyperlocal microclimate scale [31]. The backpacks housed a modular array of sensors, including a BME280 for air temperature (accuracy: ±1 °C), relative humidity (±3 %), and atmospheric pressure (±0.25 %), a CV7-V ultrasonic anemometer for wind velocity and direction, and an upward-looking (*sp*-510-SS) pyranometer for global solar radiation (±5 %). It also featured a TDS0037 NDIR CO_2 sensor and a PMS5003 laser-scattering particulate matter (PM) sensor for PM1.0, PM2.5, and PM10. A NEO-M8 GPS unit was integrated to provide high-resolution spatial mapping of the collected data. The sensors were protected with custom 3D-printed ventilated enclosures, except for those requiring direct exposure, ensuring accurate measurements. The control unit, powered by a rechargeable battery with over two hours of autonomy, wirelessly transmitted real-time data of 5 s frequency to a user-friendly dashboard accessible via Wi-Fi, making the system ideal for monitoring dynamic urban environments. In addition to the above, two microphones were used to account for basic acoustic recordings such as relative sound level and frequency. Zoom H1n compact stereo recorder featuring a unidirectional condenser microphone was used in a 90° XY stereo format, ensuring directional sound capture with reduced side noise.

2.1.3. Subjective measures

Questionnaires administered at four urban settings (Points A, B, C, D) captured subjective comfort across thermal, air quality, and acoustic domains (Supplementary Materials, Table S1) Given limited sample size, subjective data served as secondary validation for physiological observations rather than primary analytical endpoints. Perceived adaptive capacity was assessed through standardized inquiry: "In case you would experience exhaustion from extreme climate conditions, how accessible would you find suitable environments for thermal relief?" (5-point Likert scale: not at all (0) to fully accessible (+5)). This metric quantified participants' perception of available thermal refuge options, tested for correlation with measured physiological stress responses, and provided insight into psychological modulation of physiological adaptation (perceived control).

2.2. Data processing

Universal Thermal Climate Index (UTCI) calculations were performed using the wearable sensor data combining air temperature, relative humidity, solar radiation and wind speed and processed in Python via the 'pythermalcomfort' library [38] with a metabolic rate of 1.3 met (light walking activity) and clothing insulation of 0.36 clo (standard for shorts and t-shirt attire) [39].

The relative sound level in dBFS was calculated using the root mean square (RMS) of the audio signal, which measures its effective amplitude [40]. This value was then converted into a logarithmic scale relative to the maximum possible amplitude of the digital audio. A higher RMS value results in a dBFS level closer to zero, indicating a louder sound, while lower RMS values produce more negative dBFS levels, representing quieter sounds. This approach provides a standardized way to assess and compare sound intensity in digital audio analysis. To enhance the reliability of the analysis, data preprocessing includes outlier removal and signal smoothing. Outliers are identified using the Interquartile Range (IQR) method, where values falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are considered extreme and excluded. This procedure prevents anomalous values from skewing the analysis, ensuring a more accurate representation of the underlying trends. Next, time-series data is smoothed using a low-pass Butterworth filter to reduce high-frequency noise, with a carefully chosen cutoff frequency relative to the Nyquist frequency to preserve relevant trends [41]. This filtering is crucial for accurate relevant peak detection and subsequent

modelling. Peaks in the data, which signify environmental stressors and physiological responses, are then identified using prominence-based algorithm [42] that detect significant changes in signal amplitude. The detection process is governed by two key parameters: prominence, which defines the minimum relative importance of a peak, and distance, which sets the minimum time interval between successive peaks to avoid redundant detections. This step ensures that only meaningful peaks, corresponding to significant environmental or physiological events, are included in the analysis.

2.3. Correlation analysis

Initial correlation analysis quantified bivariate relationships between physiological indicators (BPM, EDA) and environmental parameters across thermal, acoustic, and air quality domains. Pearson correlation assessed linear relationships for normally distributed variables, while Spearman rank correlation captured monotonic associations for non-normal distributions [43]. This exploratory analysis addressed two objectives: (1) establishing baseline environmental-physiological associations, and (2) quantifying inter-individual variability to justify subsequent resilience modeling. All correlations were evaluated for statistical significance ($p < 0.05$), with $|r| > 0.2$ serving as an exploratory screening threshold to identify potential relationships warranting investigation through subsequent generalized logistic growth modeling, consistent with similar individual-level studies reporting $r = 0.15–0.20$ as meaningful [44]. This liberal threshold accounts for expected weak correlations in individual-level physiological data due to high inter-individual variability and complex regulatory mechanisms. The logistic growth models (Section 2.4) and clustering analysis (Section 2.5) represent the primary analytical approaches where relationship strength is properly quantified through R^2 and validated through statistical testing.

2.4. Logistic growth model for resilience dynamics

By combining signal processing techniques, peak detection, and nonlinear modelng, this study analyzes the adaptive physiological responses of each participant to urban environmental stressors. The generalized logistic growth function was selected to operationalize Schweiker's theoretical resilience framework [26] through three key advantages over alternative approaches. We employed the generalized logistic growth function [45,46] because it captures nonlinear threshold dynamics in physiological adaptation while providing mechanistically interpretable parameters (B = sensitivity, M = threshold, ν = recovery modulation) that can be extended to multi-domain stressor interactions unlike linear models (Fig. 3). Critically, the model enables multi-domain extension that preserves individual parameter interpretation while revealing interaction effects, allowing quantification of how thermal stress modulates air quality sensitivity, or how acoustic load influences thermal adaptation capacity, through coupled environmental terms rather than forcing additive assumptions. This mathematical framework, originally developed by Richards in 1959 to model plant growth rates and later widely applied in ecological modelling [46], operationalizes parameters that can be derived from empirical physiological time series, compared across individuals to quantify inter-personal resilience differences, and aggregated to neighborhood scales to map spatial patterns in collective adaptation capacity. Specifically, with particular emphasis on how fluctuations in air temperature, air quality, and noise levels affect BPM and EDA [47,48], the human response to increases or decreases in environmental stress can be modeled to reveal how multiple stressors simultaneously affect distinct physiological systems such as thermoregulation and autonomic stress response, which interact synergistically rather than independently.

To model the relationship between a single environmental stressor and physiological responses, a generalized logistic growth function as mentioned by Schweiker [26] is employed. This model captures the nonlinear dynamics of human adaptation to stress and is expressed as:

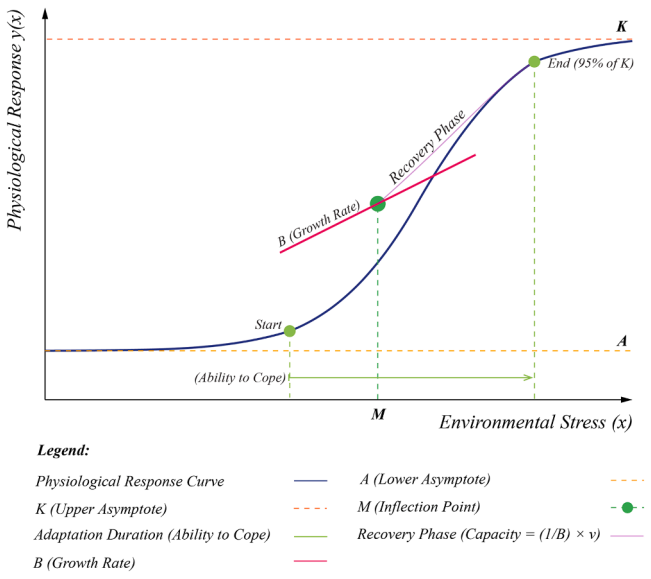


Fig. 3. Generalized Logistic Growth Model of Human Physiological Resilience. The diagram illustrates how physiological responses (y-axis) change with environmental stressors (x-axis), characterized by parameters A (baseline), K (upper limit), B (growth rate, where $1/B$ = toughness), M (inflection point), and ν (shape parameter). Recovery capacity (1/B) × ν) and adaptation duration are key resilience metrics quantifying the system's ability to respond and return to equilibrium.

$$y(x) = A + \frac{(K - A)}{(1 + e^{-B\,(x\,-\,M)})^{\frac{1}{\nu}}}$$

Where: y(x) is the physiological response as a function of environmental stress; A is the lower asymptote, representing the baseline response under normal conditions; K is the upper asymptote, representing the physiological limit; B is the growth rate, quantifying sensitivity to changes in environmental stress; M is the inflection point, indicating the stress level at which the response is most pronounced; and ν is a shape parameter controlling the steepness and symmetry of the response curve.

To account for multi-domain relationships across multiple environmental domains [49,50] the general logistic growth function was modified for the second part of this analysis as follows:

$$y(x_1, x_2) = A + \frac{(K - A)}{(1 + Q \cdot e^{-(B_1(x_1 - M_1) + B_2(x_2 - M_2))})^{\frac{1}{\nu}}}$$

Where: y(x1,x2) represents physiological response as a function of two environmental variables pertaining to different environmental domains, e.g., CO_2 concentration (x_1) and sound level (x_2); A and K are the lower and upper limits of the physiological response; B_1 and B_2 are growth rates relative to each environmental stressor; M_1 and M_2 are inflection points (threshold values where significant physiological signal changes occur, for each environmental stressor); and Q is an asymmetry factor. The fitted parameters of the function such as A (lower asymptote), K (upper asymptote), B (growth rate), M (inflection point), and ν (shape parameter), are used to derive three key resilience metrics: toughness (T), capacity to recover (RC), and ability to cope (AC) (Table 1). The fitting parameters and resilience metrics were first calculated for a reference participant to illustrate the general logistic growth model. To evaluate collective physiological resilience, the study then applied the same analytical approach across multiple participants. This process allowed for a systematic assessment of the multi-domain model's performance and the categorization of resilience parameters into distinct

Table 1
Resilience Metrics calculated from the fitting parameters of the general logistic growth function for both single and multi-domain approach.

Resilience Metric	Definition	Formula
Toughness (T)	Toughness quantifies the resistance of the system to change and is inversely proportional to the growth rate (B).	$\tau = 1 / B$
Recovery Capacity (RC)	Recovery capacity combines resistance (τ) with the shape parameter (ν) to describe the ability of the system to recover from stress.	$RC = \nu\,\tau = \nu / B$
Adaptation Duration or Ability to Cope (AC)	The duration of the ability to cope is determined as the time difference ($\Delta t = t_{95}$ % - tpeak) between the occurrence of the first peak in physiological response and the point where the response reaches 95% of the asymptotic range between the lower asymptote (A) and upper asymptote (K). This approach assumes the complete development of ability to cope, which was ensured in our experimental design as participants' behaviour was controlled, preventing adaptive actions such as hydration or clothing adjustments that might otherwise truncate the thermal stress response before reaching upper asymptotic levels.	$\Delta t = t_{95\ \%} - tpeak$

groups.

These resilience metrics (Table 1) provide mechanistic links between environmental design and physiological outcomes, operationalizing Schweiker's [26] three-phase resilience framework. Toughness ($T = 1/B$) quantifies resistance to initial environmental stress, describing how slowly physiological responses escalate as stressors increase. Higher toughness values indicate individuals who maintain baseline function despite rising thermal stress requiring fewer immediate interventions. However, excessive toughness may delay recognition of accumulating strain, particularly in vulnerable populations who may report thermal conditions as acceptable despite associated health risks [51]. Recovery Capacity ($RC = \nu/B$) measures the speed of return to homeostatic equilibrium after peak stress exposure ends. This metric directly informs urban cooling infrastructure design: populations with low recovery capacity require more frequent spatial access to thermal refuges, as they cannot efficiently restore physiological equilibrium between exposures. Adaptation Duration ($AC = t_{95}$ % - tpeak) quantifies the time required for physiological stabilization after stress peaks, representing what Schweiker describes as the "ability to cope" phase. This metric establishes minimum exposure thresholds for intervention effectiveness. If cool spots or green spaces provide relief periods shorter than an individual's adaptation duration, the physiological system cannot complete its recovery process, leaving residual strain that accumulates across repeated exposures. Critically, these person-specific metrics enable identification of low-resilience individuals who may be missed by demographic vulnerability indices, while simultaneously aggregating to neighborhood scales for spatial mapping of collective adaptive capacity [52].

Single-domain models (5 parameters: A, K, B, M, ν) tested seven environmental predictors independently: UTCI, CO_2, average PM, overall sound level (dBFS), and three frequency-specific acoustic measures (low-frequency: <500 Hz representing traffic and mechanical noise; mid-frequency: 500–2000 Hz capturing human voices and urban activity; high-frequency: >2000 Hz indicating birds, wind, and high-pitched mechanical sounds). Multi-domain models (8 parameters: A, K, B_1, B_2, M_1, M_2, Q, ν) tested eleven pairwise combinations: UTCI paired with CO_2, PM, and each acoustic frequency band; CO_2 paired with each acoustic frequency band; and PM paired with each acoustic frequency band. Each configuration was fitted separately for BPM and EDA across all participants, with only models achieving $R^2 > 0.20$ reported in Tables 4–5.

Generalized logistic growth models were fitted as within-participant models to account for repeated measurements collected during each individual's transect walk (multiple time points per participant across varying environmental exposures). Models were then fitted independently for each participant using nonlinear least squares optimization. This participant-specific approach enabled individual heterogeneity quantification as required by the theoretical framework, while the small sample size ($n = 8$) precluded reliable mixed-effects modeling across participants. Model performance was assessed through adjusted R^2 accounting for differing predictor numbers between single-domain (5 parameters) and multi-domain (8 parameters) configurations. Leave-one-participant-out (LOPO) validation ensured findings were not driven by outliers. We recalculated R^2 values and resilience metrics after systematically excluding each participant, with stability assessed through the range of LOPO mean predictions (<0.15 considered robust) [53]. Coefficient of variation (CV) was calculated for full sample and each LOPO iteration to confirm heterogeneity (CV>50%) persisted across participant subsets. R^2 values represent within-participant model fit, not population-level variance. This individual-specific approach was required to: (1) quantify individual heterogeneity (our primary research question), (2) extract mechanistically interpretable resilience metrics (Toughness = 1/B and Recovery Capacity = ν/B) from participant-specific parameters, and (3) avoid unstable nonlinear mixed-effects estimation with limited sample size ($n = 8$). The within-subject modeling inherently accounts for repeated measures clustering.

2.5. Individual spatial resilience analysis and adaptation zone classification

We performed K-means clustering on T/RC/R^2-weighted environmental features to identify spatial resilience zones, as our normalized composite features exhibit compact distributions suitable for this approach rather than requiring density-based methods. Each environmental feature was assigned a weight proportional to its temporal contribution (T), recovery coupling (RC), and explained variance (R^2), derived from prior model evaluation. For each domain (air, acoustic, thermal), normalized composite weights were computed as:

$$w_i = \frac{\left(T_i + RC_i + R_i^2\right)}{\sum\left(T + RC + R^2\right)}$$

Where: T_i represents the temporal persistence of the environmental signal; RC_i denotes its coupling with recovery capacity; and R^2_i quantifies its explanatory strength within the environmental–physiological model. The numerator therefore reflects the combined influence of temporal relevance and physiological linkage, while normalization by the summed contributions across all domains ensures that weights sum to one. These weights were then used to scale each standardized environmental variable, producing empirically grounded composite multi-domain features prior to clustering. All variables were z-score normalized prior to analysis. Cluster validity was evaluated across $k = 2$–9 using silhouette coefficient (measuring within-cluster cohesion versus between-cluster separation) and Calinski–Harabasz index (ratio of between-cluster to within-cluster variance). The optimal k was selected balancing compactness and interpretability. Post-clustering stability was assessed through bootstrap resampling: 1000 iterations of sampling with replacement, refitting K-means models, and calculating the percentage of iterations each observation was assigned to its modal cluster. Post-clustering, the resilience index, a normalized indicator of the adaptation metrics calculated (Section 2.4) and the average physiological stress deriving from HR and EDA data entries were compared across zones using one-way ANOVA. Zones were then ranked by mean resilience to represent a gradient from optimal adaptive environments to low-capacity, personalized stress-prone zones.

3. Results

The design of the transect route was subsequently validated using subjective comfort vote data. Point A was characterized as warm and humid based on thermal sensation metrics and was perceived as slightly to extremely uncomfortable in terms of thermal and air quality conditions by 87.5% of participants. In contrast, Point B exhibited similar warm and humid conditions but was reported as thermally more uncomfortable by 62.5% of participants. During each session, thermal conditions were monitored through UTCI readings, ranging from 24.84 ± 0.88 °C to 30.20 ± 0.67 °C. Physiological responses were tracked via heart rate (BPM: 86.93 ± 13.24 to 100.61 ± 16.64 bpm) and EDA (2.82 ± 1.04 to 47.08 ± 17.91 μS). Environmental quality parameters included PM measurements at three levels (PM1.0: 4.25–13.70 μg/m^3, PM2.5: 7.02–16.69 μg/m^3, PM10: 6.91–17.28 μg/m^3), carbon dioxide levels (354.39–438.31 ppm), and solar radiation via pyranometer readings (0.04–1704.91 W/m^2). Acoustic environment was characterized through sound level measurements (−59.37 to −61.36 dBFS) and frequency analysis across low, mid, and high ranges. Local microclimate conditions comprised temperature (25.46–30.04 °C) and relative humidity (49.17–52.18 %RH) readings.

3.1. Preliminary correlation analysis

Correlation analysis was conducted for each participant individually, examining relationships between target variables (UTCI, BPM, and EDA) and environmental parameters across all domains (thermal, acoustic, air quality), one physiological parameter (skin temperature), and one subjective measure (perceived adaptation access). The analysis demonstrated varying degrees of association with environmental factors, suggesting predominantly individualized response patterns with a few consistent correlations observed across multiple participants (Fig. 4). Specifically, as expected, UTCI exhibited the most consistent relationship with air temperature, with correlation coefficients ranging from $r =$ 0.53 to $r = 0.98$ across all participants. However, other environmental parameters demonstrated notable variability in their associations. Humidity exhibited inconsistent correlations, ranging from moderately negative ($r = -0.49$) to slightly positive ($r = 0.22$), suggesting that contextual factors may modulate its influence on UTCI. Particularly noteworthy were the divergent correlations with air quality parameters. While some participants UTCI showed strong positive correlations with PM (e.g., Participant 3: $r \approx 0.60$–0.66), others demonstrated moderate negative correlations (e.g., Participant 4: $r \approx -0.31$ to -0.35). BPM correlations revealed marked individual differences in physiological responses to environmental stimuli.

Relative sound levels demonstrated reliably negative associations with BPM ($r = -0.28$ to -0.45). Moreover, the relationship between air quality metrics and heart rate varied substantially between participants, with both positive and negative correlations observed. EDA demonstrated the highest degree of inter-individual variability among the measured physiological parameters. Temperature correlations ranged from moderately positive ($r = 0.42$) to negative ($r = -0.29$). Humidity showed relatively consistent correlations with EDA. Particulate matter correlations were inconsistent, indicating that air quality may not have a uniform influence on electrodermal activity across individuals. A consistent negative correlation emerges between adaptation access and various physiological and environmental variables. Specifically, participants who perceived easier access to spaces for heat stress relief, assessed at different points of interest (A, B, C, D), tended to exhibit lower BPM. The strongest negative correlations are observed with BPM, averaging −0.44, with this effect being most pronounced in 4 out of 8 participants (Fig. 4). This trend suggests that improved adaptation access may help reduce physiological responses, as indicated by negative correlations with BPM (−0.29 to −0.44) and EDA (−0.44 to −0.20). These physiological changes likely reflect both direct environmental effects and significant psychological components, including higher perceived levels of control and reduced perceived heat risk when adaptation options such as accessible heat shelters are available [54].

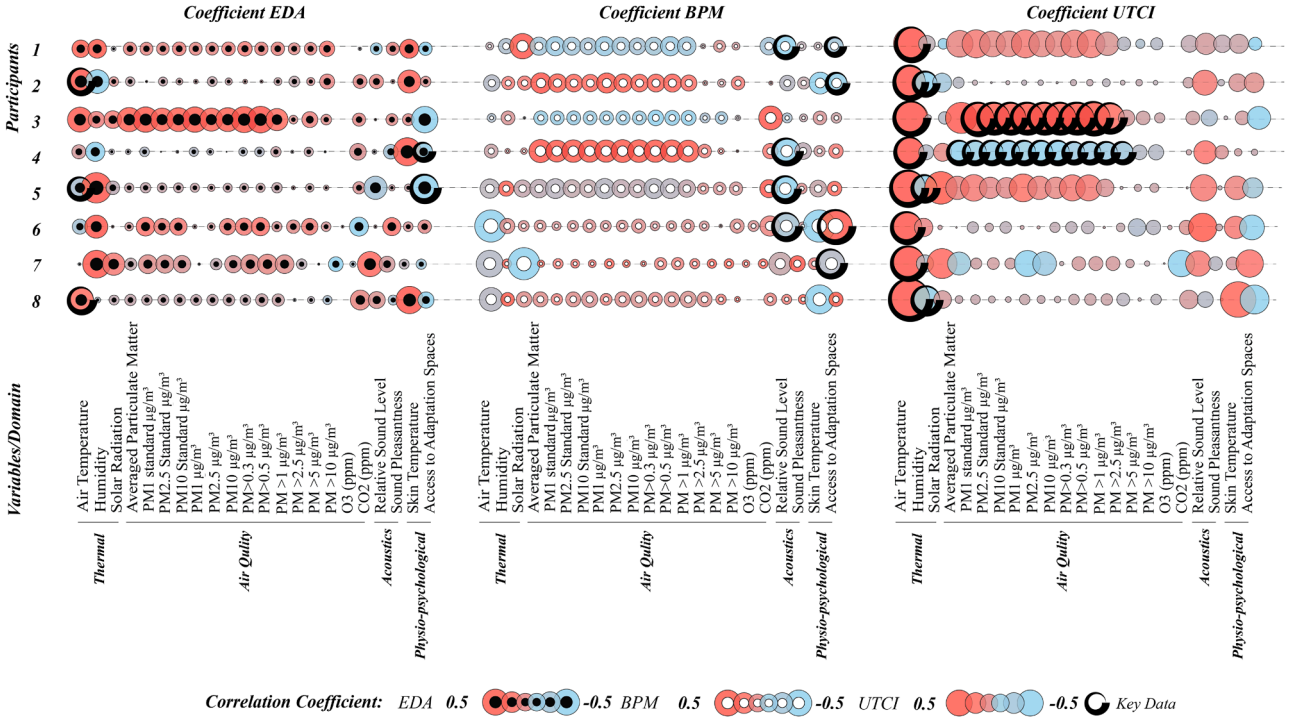


Fig. 4. Average and individual correlation coefficients of physiological responses (BPM and EDA) and UTCI to multiple environmental stressors during the experimental campaign. Annotated correlations indicate key values discussed in text: UTCI-air temperature, UTCI-humidity, UTCI-particulate matter for Participant 3 and Participant 4, BPM-sound level, BPM-adaptation access for 4 participants, EDA-air temperature, and EDA-adaptation access. Bubble size represents correlation magnitude; color indicates direction (red = positive, blue = negative) and marked are the key data discussed within Section 4.1.

This variability suggests physiological stress is modulated by factors beyond single-domain conditions and demographic categories alone. This highlights the limitations of current approaches and highlights the need for a more comprehensive framework to accurately quantify human vulnerability to the combined effects of multiple environmental stressors.

3.2. Single domain approach

Taking as an example a random reference participant, in single-domain analysis (Fig. 5), environmental variables such as CO_2 concentration, sound levels, and average particulate matter concentration display varying degrees of impact on heart rate, as reflected by toughness, recovery capacity, and adaptation duration summarized in Table 2. Single-domain analysis revealed variable resilience patterns: CO_2 showed moderate toughness (0.212) with good recovery (0.794), while PM exhibited lower toughness (0.129) with no recovery capacity. Acoustic stressors demonstrated frequency-dependent responses (Table 2). These results indicate that different environmental factors affect physiological responses through distinct resilience patterns. Mid and high-frequency sounds have the most consistent impact on this participant ($R^2 = 0.13$ and 0.18 respectively), suggesting that auditory stimuli may be a key determinant of physiological stress. Notably, UTCI, low-frequency sounds and relative sound level measurements did not produce reliable resilience models for this participant, indicating a lack of direct physiological response to these variables using the single-domain approach.

3.3. Multi-domain approach

Taking the same reference participant as an example for our multi-domain model (Table 3), when UTCI is paired with CO_2 concentration, BPM response shows relatively high toughness (0.294) but zero recovery capacity, suggesting that while this combination resists initial changes in heart rate, it severely compromises the system's ability to recover. This pairing represents a form of multi-domain overload where the thermoregulatory system maintains initial stability but cannot adapt over time. In contrast, when UTCI interacts with average low frequency sounds, we observe a much lower toughness (0.098) but significantly higher recovery capacity (1.498), indicating greater physiological sensitivity but enhanced resilience over time. The combination of CO_2 with acoustic stressors reveals also interesting patterns. When CO_2 is paired with high-frequency sounds, the model predicts moderate toughness (0.099) and high recovery capacity (1.469), suggesting that this multi-domain interaction initially challenges physiological stability but allows for robust recovery. However, with mid-frequency sounds, the same CO_2 levels yield a lower recovery capacity (0.278), indicating frequency-specific physiological responses that would be missed in single-domain analyses. Particularly noteworthy is the average PM interaction with high-frequency sounds, which shows the highest toughness (0.259) and recovery capacity (3.348) values among all combinations. This finding indicates that this pairing creates a more resilient physiological response compared to other environmental combinations, potentially indicating an adaptive combined effect unique to this pairing.

Surface plots from these models reveal how heart rate responds to combined environmental stressors (Fig. 6). The R^2 values vary considerably (0.036–0.345), indicating that different environmental pairings affect physiology in distinct ways. UTCI emerges as the dominant modulator across most interactions. The strongest predictor of physiological response is the UTCI–particulate matter pairing ($R^2 = 0.345$). Temperature–PM interactions ($R^2 = 0.34$) show poor initial tolerance ($T = 0.010$) but moderate recovery (RC = 0.160), suggesting adaptive capacity despite vulnerability. In contrast, temperature–CO_2 pairing demonstrates strong initial resistance ($T = 0.294$) but complete recovery failure, indicating a brittle response under prolonged exposure. Interactions involving high-frequency sound show notably robust recovery: CO_2–sound pairings achieve RC = 1.469 with $T = 0.099$, while PM–sound interactions produce the most resilient profile overall ($T = 0.259$; RC = 3.348).

The relatively high R^2 values for CO_2 paired with sound frequencies (0.238, 0.210, 0.117) compared to UTCI with sound frequencies (0.036, 0.077, 0.191) suggest that CO_2 may have more consistent interactive effects with acoustic stressors than thermal conditions do. This profile suggests the participant may function best in environments where PM levels and acoustic factors are carefully managed, as these combinations trigger their most resilient physiological responses, while showing greater vulnerability to thermal-CO_2 interactions despite initial

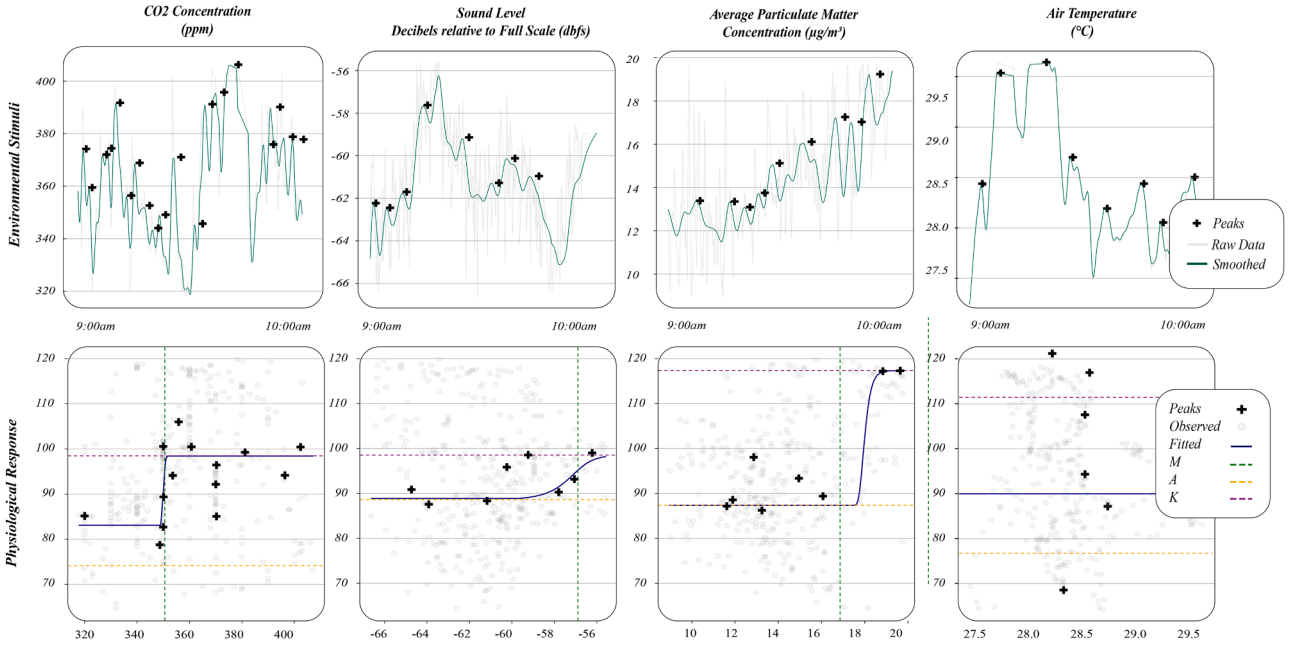


Fig. 5. Resilience Analysis of Physiological Response for the reference participant. Time series (top) and logistic growth modelling (bottom) of BPM responses to environmental stimuli. Peaks indicate stress events.

Table 2
Resilience metrics – single domain analysis for the reference participant.

Variable	T	RC	AC	R^2	A	K	B	M	ν
UTCI	-	-	-	-	-	-	-	-	-
CO_2	0.21	0.79	39	0.18	11.44	16.42	4.72	357.31	3.75
PM	0.12	0	59.5	0.06	14.40	16.80	7.77	17.90	0
Sound Level	-	-	-	-	-	-	-	-	-
Low freq Sounds	-	-	-	-	-	-	-	-	-
Mid freq Sounds	0.13	0.73	42.83	0.12	18.34	14.36	7.46	−1.52	5.47
Hight freq Sounds	0.18	0.45	42.50	0.16	16.68	13.78	5.44	−25.49	2.47

Note: The relationship between various environmental stressors and human resilience, as measured by Toughness (T), Recovery Capacity (RC), and Ability to Cope (AC) in relation to the selected participant' BPM. Each variable's resilience metrics are summarized alongside the R^2 value, indicating the goodness of fit for the model, and the fitting parameters (A, K, B, M, ν), which describe the underlying mathematical model. Note: Dashes (-) indicate where values were not available or were extremely large, indicating overfitting or lack of direct response.

Table 3
Resilience metrics – multi-domain analysis for the reference participant.

Domains	T	RC	AC	R^2	A	K	B1	B2	M1	M2	Q	Nu
UTCI vs CO2	0.29	0	61.33	0.19	-	87.62	3.396	0	0.38	0.43	0	0
UTCI vs PM	0.01	0.16	61.33	0.34	91.01	114.62	0	99.94	0	0.73	15.96	0.05
UTCI vs Low f Sounds	0.09	1.49	61.33	0.03	92.60	102.29	0	10.25	0.53	0.62	15.36	2.09
UTCI vs Mid f Sounds	0.01	0.09	61.33	0.07	92.16	100.91	0	52.21	0.51	0.37	4.90	0.01
UTCI vs High f Sounds	0.29	0	61.33	0.19	-	86.99	3.19	0.23	0.29	0.33	0	0.02
CO2 vs Low f Sounds	0.01	0.75	61.16	0.23	91.28	116.56	0	54.95	0.69	0.47	41.25	0
CO2 vs Mid f Sounds	0.07	0.27	61.16	0.21	82.27	121.10	0.90	13.03	0.97	0.79	3.87	3.17
CO2 vs High f Sounds	0.25	3.34	59.66	0.13	88.07	138.62	0	3.85	0.52	0.53	12.90	1.53
PM vs avg_low_freq	0.19	2.42	59.66	0.09	87.04	121.31	0	5.22	0.56	0.57	12.68	1.98
PM vs avg_mid_freq	0.21	1.91	59.66	0.16	82.10	122.87	0	4.56	0.55	0.56	8.71	2.48
PM vs avg_high_freq	0.25	3.34	59.66	0.13	88.07	138.62	0n	3.85	0.52	0.53	12.90	1.53

Note: The relationship between various environmental stressors and human resilience, as measured by Toughness (T), Recovery Capacity (RC), and Ability to Cope (AC) in relation to the selected participants' BPM (beats per minute). Each variable's resilience metrics are summarized alongside the R-squared value, indicating the goodness of fit for the model, and the fitting parameters (A, K, B, M, ν), which describe the underlying mathematical model.

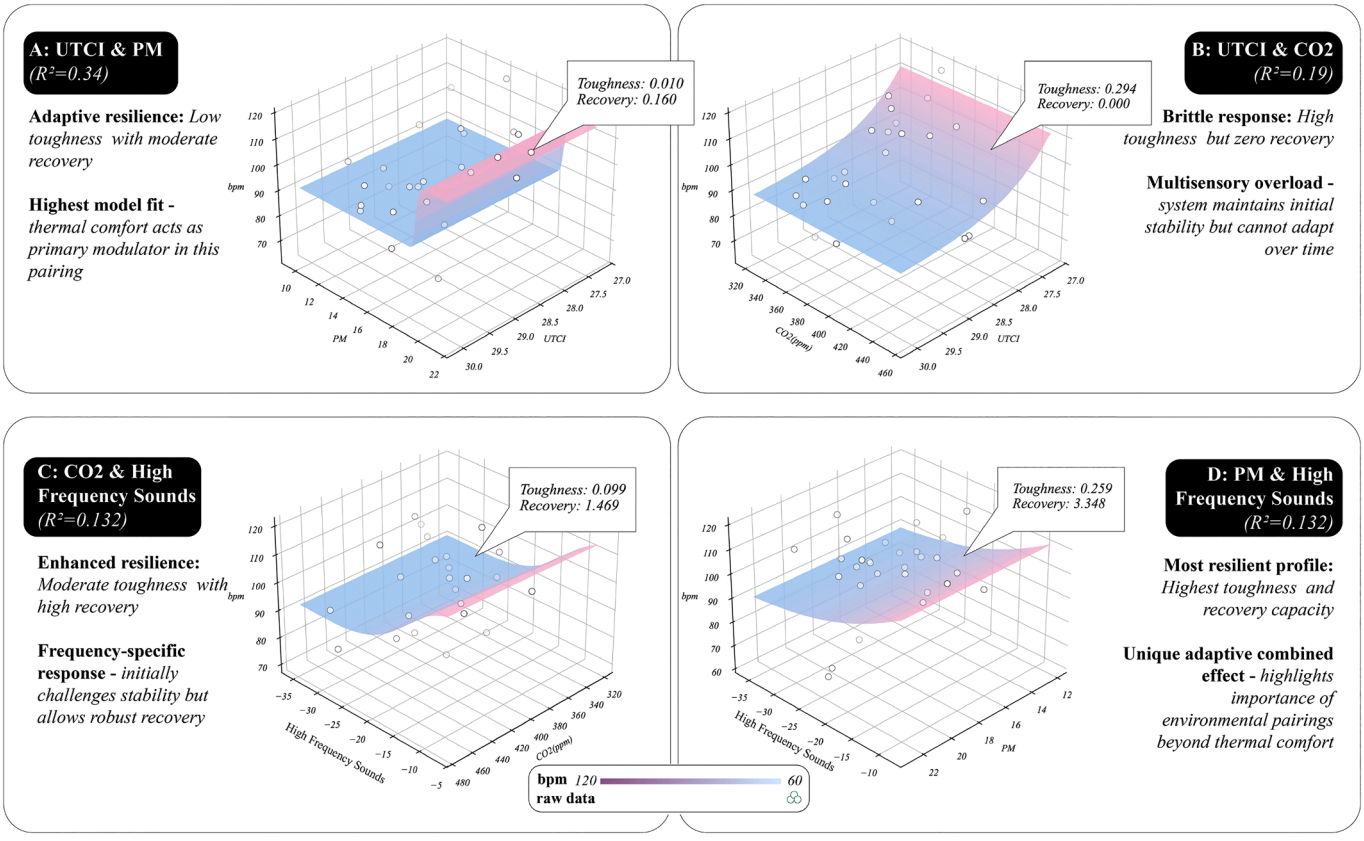
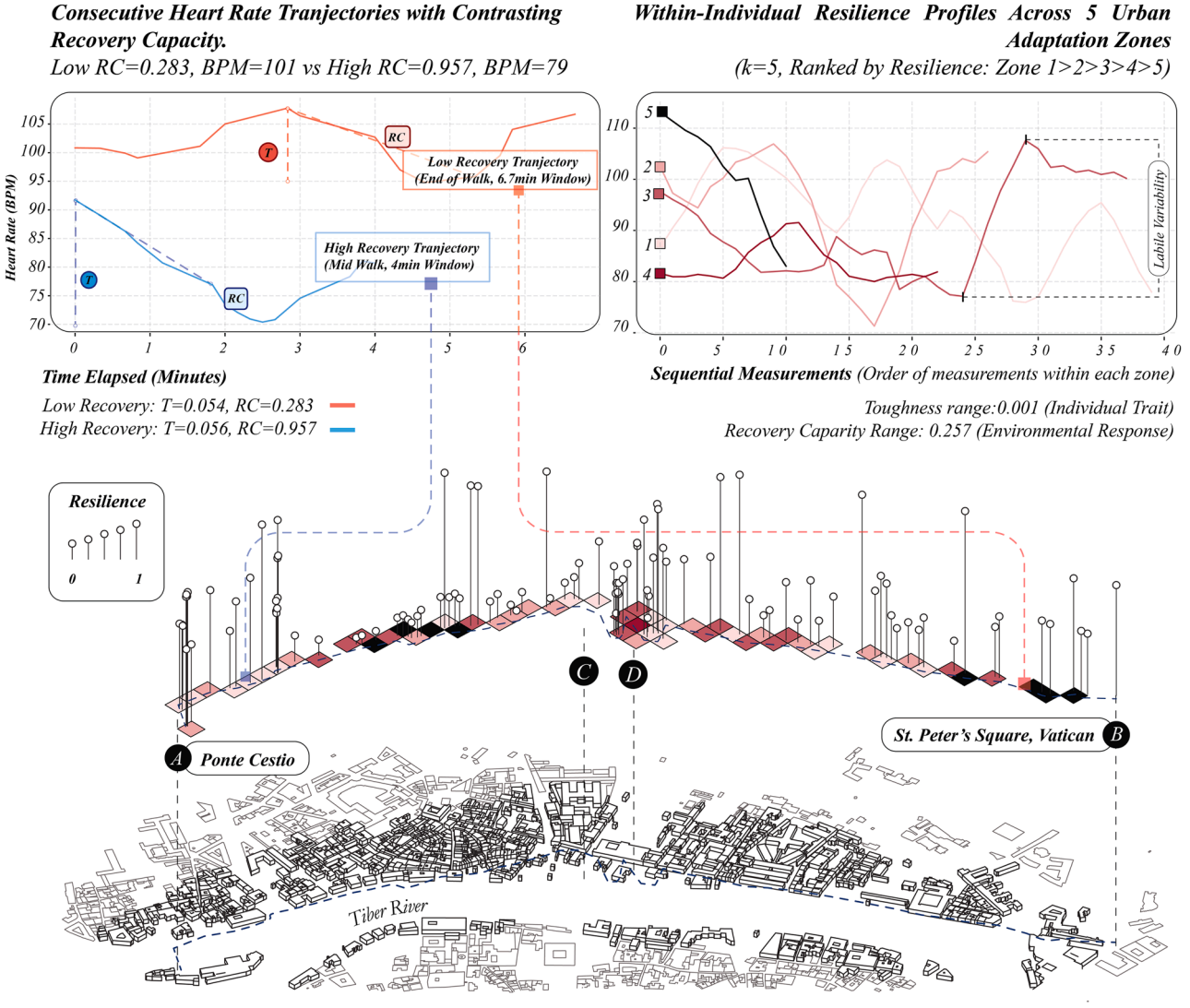


Fig. 6. Three-dimensional response surfaces showing heart rate (BPM) as a function of paired environmental stressors for the reference participant. Surfaces fitted using generalized logistic growth models; scatter points represent observed data.

resistance to these stressors. This finding highlights the importance of considering also environmental pairings beyond thermal comfort in multi-domain comfort assessment.

3.4. Personalized adaptation zone classification

Using 164 multi-domain observations, we applied k-means clustering to five T/RC/R^2-weighted environmental features (weighted_air, weighted_acoustic, weighted_thermal, UTCI × PM, CO_2 × low-freq, CO_2 × mid-freq), integrating each domain's temporal persistence (T), recovery coupling (RC), and explanatory strength (R^2). Cluster validation confirmed a stable five-zone solution (Silhouette = 0.42; Calinski–Harabasz = 60.4), ranked by resilience from Zone 1 (R = 0.657, balanced conditions: CO_2=426 ppm, UTCI=30.1 °C) to Zone 5 (R = 0.405, acoustic stress, mean BPM=101.9). Fig. 7 visualizes this multi-scale resilience framework. The top-left panel shows contrasting heart rate trajectories with stable toughness ($T \approx 0.055$) but variable recovery capacity (RC=0.283 vs 0.957), validating T as an individual trait and RC as environmentally sensitive. Red trajectory (Point D, Zone 5) shows dysregulated elevation; blue trajectory (Point C) shows homeostatic recovery. The top-right panel confirms within-individual variation (T range=0.001, RC range=0.257) across all zones. The bottom panel presents the 3D spatial map with zones ranked by resilience (1 > 2 > 3 > 4 > 5). Zone 5′s elevated heart rate reflects dysregulated stress responses where resilience measures regulatory control and recovery efficiency, not activation magnitude. Within-individual comparison validated the person-environment model: toughness remained stable (range=0.001), while recovery capacity varied substantially (range=0.257). Two



Cluster	Resilience	Toughness	Recovery	BPM	UTCI	CO_2	AQI	Low-freq	High-freq	Environmental Notes
1	0.657 ± 0.174	0.071	0.598	96.1 ± 11.9	30.1	426	11.3	11.7	−24.7	Balanced multi-domain conditions, optimal resilience
2	0.590 ± 0.192	0.056	0.635	94.6 ± 14.9	29.9	417	10.6	13.1	−24.1	Moderate thermal load, high recovery capacity
3	0.567 ± 0.189	0.055	0.606	90.2 ± 12.5	30.4	420	11.4	12.1	−25.0	Low acoustic stress, stable air quality
4	0.464 ± 0.090	0.056	0.482	83.5 ± 5.1	30.4	484	13.4	4.3	−31.2	Thermal-air quality interaction, reduced recovery
5	0.405 ± 0.110	0.055	0.664	101.9 ±	29.2	411	11.8	19.4	−12.9	Acoustic-dominated stress, dysregulated response

Fig. 7. Environmental and physiological resilience across clustered zones (k = 5). Multi-scale resilience framework integrating spatial and temporal dynamics.

Table 4
An overview of the influence of environmental domain pairs on the resilience metrics across all the participants tested.

Domain Pair	Participants with Significant Model	Average R^2 (±SD)	T (Avg ± SD)	RC (Avg ± SD)	AC (Avg ± SD)
UTCI & PM	4/8	0.40 ± 0.25	0.01 ± 0	0.06 ± 0.07	62.92 ± 11.65
UTCI & CO_2	4/8	0.20 ± 0.18	0.12 ± 0.13	0 ± 0	62.92 ± 10.97
CO_2 & High Freq Sounds	6/8	0.32 ± 0.13	0.05 ± 0.03	0.36 ± 0.55	63.40 ± 10.26
PM & High Freq Sounds	8/8	0.22 ± 0.09	0.07 ± 0.08	0.59 ± 1.19	61.71 ± 9.58
CO_2 & Low Freq Sounds	6/8	0.26 ± 0.13	0.03 ± 0.02	0.21 ± 0.27	63.20 ± 9.89
CO_2 & Mid Freq Sounds	6/8	0.25 ± 0.13	0.04 ± 0.03	0.18 ± 0.16	63.20 ± 9.89
PM & Low Freq Sounds	7/8	0.23 ± 0.15	0.05 ± 0.06	0.47 ± 0.90	62.70 ± 9.34

Note: *The pairs depicted in this table are those with R^2 values exceeding 0.2, representing a minimum threshold of statistical importance in explaining variance in BPM regulation.*

Table 5
An overview of the influence of environmental domain pairs on the resilience metrics across all the participants tested.

Domain Pair	Participants with Significant Model	Average R^2 (±SD)	T (Avg ± SD)	RC (Avg ± SD)	AC (Avg ± SD)
UTCI & PM	5/8	0.43 ± 0.20	0.05 ± 0.04	0.16 ± 0.20	61.66 ± 11.05
UTCI & CO_2	5/8	0.32 ± 0.22	0.05 ± 0.04	0.14 ± 0.15	61.66 ± 11.05
CO_2 & High Freq Sounds	8/8	0.21 ± 0.12	0.06 ± 0.08	0.09 ± 0.23	63.62 ± 9.75
PM & High Freq Sounds	8/8	0.21 ± 0.17	0.03 ± 0.01	0.14 ± 0.25	63.12 ± 9.04
CO_2 & Low Freq Sounds	8/8	0.25 ± 0.15	0.09 ± 0.14	0.25 ± 0.59	63.62 ± 9.75
CO_2 & Mid Freq Sounds	8/8	0.25 ± 0.14	0.05 ± 0.04	0.12 ± 0.19	63.62 ± 9.75
PM & Mid Freq Sounds	8/8	0.20 ± 0.21	0.02 ± 0.01	0.13 ± 0.23	63.12 ± 9.04

Note: The pairs depicted in this table are those with R^2 values exceeding 0.2, representing a minimum threshold of statistical importance in explaining variance on EDA.

consecutive trajectories illustrate this: high-recovery (Point C, RC=0.957) showed efficient baseline return (90→70 BPM), while low-recovery (Point D, Zone 5, RC=0.283) showed sustained elevation (100–107 BPM) despite similar toughness. Environmental quality modulates adaptive responses through recovery mechanisms, with urban microenvironments directly shaping physiological regulation capacity.

3.5. Multi-domain collective resilience

The following section presents the results of this multi-domain assessment, highlighting patterns and differences in physiological responses among all the participants aiming to summarize the resilience metrics collectively. Findings reveal that the combination of PM concentration with high frequency sounds shows the most consistent relationships to BPM (present in all 8/8 participants, $R^2 = 0.220$) (Table 4), while the pairing of UTCI with PM produces the highest average R^2 (0.400 across 4/8 participants). These relationships demonstrate varying resilience patterns: the UTCI-PM domain pairing shows relatively low toughness (0.016) but measurable recovery capacity (0.066), while CO_2 paired with high frequency sounds shows more moderate toughness (0.057) and higher recovery capacity (0.369) with an R^2 of 0.323 observed in 6/8 participants. This finding suggests that environmental variables influence human physiological responses through different mechanisms, with thermal domain (UTCI) playing a central moderating role in the most significant relationships.

While BPM demonstrates robust relationships with multiple environmental domains, particularly when UTCI interacts with PM ($R^2 = 0.400$) and when CO_2 interacts with sound frequencies ($R^2 = 0.323$ for high frequency), EDA exhibits even stronger associations (Table 5), especially with thermal-air quality domains. For EDA, the UTCI-PM pairing exhibits the highest average R^2 (0.434 across 5/8 participants) with moderate toughness (0.055) and recovery capacity (0.168). Similarly, the UTCI-CO_2 combination shows meaningful correlations ($R^2 = 0.328$) with comparable resilience metrics (toughness = 0.050, recovery capacity = 0.145). This differential response pattern suggests a hierarchical physiological adaptation mechanism in outdoor environments, where EDA appears particularly attuned to thermal and air quality parameters, showing higher correlation strengths than cardiovascular responses for these specific domain pairs.

Notably, both physiological measures (BPM and EDA) show consistent relationships with acoustic stressors across all participants, though with varying strengths. The sound frequency pairings with CO_2 and PM demonstrate universal significance across all participants for EDA (8/8 participants for all frequency ranges), suggesting that these environmental combinations reliably trigger electrodermal responses regardless of individual differences. The PM-low frequency sound combination shows particularly high recovery capacity for EDA (0.894), indicating robust adaptation potential for this specific environmental pairing. This dual-measure approach reveals complementary aspects of human physiological adaptation to multi-domain environmental stressors, with cardiovascular responses (BPM) showing strongest correlations to UTCI-PM interactions, while electrodermal responses (EDA) demonstrate stronger associations with both thermal-air quality and thermal-acoustic domain pairs. These findings highlight the importance of considering integrated multi-domain environmental quality assessments aligned with recent studies supported by crowdsourced data on human perception.

Leave-one-participant-out (LOPO) validation confirmed robustness of these multi-domain relationships (Supplementary Table S2). Multi-domain superiority persisted across all iterations. For heart rate responses, five of seven domain pairs (71 %) showed LOPO stability <0.15 (meaning the predictive strength remained consistent when each participant was systematically excluded, considered robust for this study), with PM-acoustic combinations demonstrating particularly robust findings (stability: 0.03–0.06) and CO_2-acoustic pairings showing consistent predictive power (stability: 0.11–0.12). EDA demonstrated higher between-participant variability, with UTCI-PM (stability: 0.18) and UTCI-CO_2 (stability: 0.19) showing moderate variability, yet both EDA domain pairs still substantially outperformed single-domain thermal models.

To justify the heterogeneity finding statistically, we ran Kruskal-Wallis tests across all 18,815 observations. Both BPM ($H = 1358.1$, $p < 0.001$) and EDA ($H = 13{,}937.7$, $p < 0.001$) showed significant differences between participants. EDA means ranged from 2.68 to 44.18 μS (CV=89.6 %), and Cohen's d reached 0.96 for BPM and 3.15 for EDA, both large effects (Table S3). Crucially, these differences persisted even when comparing participants within the same CO_2 exposure range (330–450 ppm), where individual mean BPM still spanned 89.1–100.9 bpm. This suggests the differences reflect actual physiological variability between individuals, not differences in environmental exposure. These effect sizes are consistent with literature showing that individual

variability in thermal comfort is on the order of 3°C in neutral temperature, often comparable or exceeding reported differences between demographic groups [55].

4. Discussion

4.1. Multi-Domain environmental interactions override single-stressor predictions

Vulnerability is not a static spatial pattern: it varies across individuals, across time of day, and across sensory combinations, a fact that aggregate indices obscure and that only person-scale physiological monitoring can recover. Our findings confirm the multi-domain framework's explanatory superiority established in the results. This challenges Attention Restoration Theory (ART) and Stress Recovery Theory (SRT), which traditionally emphasize visual nature exposure as the primary restoration mechanism. Our results demonstrate that UTCI functions as a moderator of other stressors rather than an independent driver, revealing multi-domain synergies absent from classical single-domain restoration frameworks. Temperature-PM interactions showed the strongest correlations (BPM: R^2=0.40; EDA: R^2=0.43), while CO_2-acoustic frequency pairings demonstrated universal electrodermal responses across all participants. These compound effects extend beyond visual-natural restoration pathways emphasized in traditional theory [21]. PM-acoustic pairings succeeded where UTCI alone failed, indicating cross-domain interactions, not singular thermal or visual drivers, determine adaptive outcomes. These findings partially align with sensory and cognitive load theories suggesting that urban overload from noise, density, and pollution does elevate stress through autonomic activation. But the patterns observed go beyond that: EDA responded more strongly to thermal-air quality interactions than cardiovascular measures did, suggesting that different physiological systems handle different environmental loads, and these loads interact synergistically rather than additively. Optimal resilience emerged from balanced multi-domain conditions (Zone 1: CO_2=426 ppm, UTCI=30.1 °C with low-to-mid frequency acoustic dominance and minimal high-frequency intrusion) rather than maximizing single factors like vegetation extent or minimizing thermal exposure alone. Tree-canopied areas with poor circulation showed elevated discomfort undermining recovery despite visual-natural qualities [56], demonstrating that greenery alone is not sufficient to ensure improved physiological outcomes. This contributes to explaining mixed evidence in existing density and greenery research [23]further framing that previous studies capturing single-domain effects (visual greenery, thermal relief) may have missed the compound thermal-air quality-acoustic interactions that actually determine physiological outcomes, leading to inconsistent restoration findings across contexts. Hong and Jeon [7] showed that pleasantness and visual quality together contributed similarly to perceived environmental harmony (standardized path coefficients of 0.37 and 0.34 respectively), yet climate adaptation planning continues to optimize what sensors can measure remotely rather than validating what bodies experience locally. Our results confirm that landscape-mediated comfort emerges from integrated thermal-air quality-acoustic conditions shaped by vegetation placement, spatial configuration, and material selection, not temperature reduction, greenery extent, or density alone [11,57].

4.2. Individual heterogeneity

In this pilot sample, individuals with similar observable characteristics (age 28–38, healthy adults, similar educational backgrounds) exhibited markedly different physiological responses to identical microclimates. Kruskal-Wallis tests suggest highly significant inter-individual differences in both BPM and EDA across all 18,815 observations, with large effect sizes persisting within identical CO_2 exposure ranges (Section 3.5, Table S3), consistent with Wang et al. and their review on inter-individual thermal comfort variability [55]. The same temperature-PM combination producing high toughness ($T = 0.294$) but zero recovery in one participant generated low toughness ($T = 0.010$) with moderate recovery (RC=0.160) in another. To check whether findings depend on a single day's conditions, we stratified analyses by session (Table S4). The two days differed slightly. Day 2 was approximately 2.7 °C cooler with lower PM2.5 but CO_2 exposure was comparable and both days fell within the same UTCI range. Physiological heterogeneity was statistically significant within each day independently, confirming the finding is not an artifact of either day's specific conditions. While this small-sample observation cannot validate whether demographic categorization fails broadly, it demonstrates that at least within this homogeneous group, population-averaged assumptions did not predict individual stress responses, supporting the rationale for personalized physiological monitoring in larger validation studies. While GIS-based modeling remains essential for spatial patterns [16,17], it cannot predict individual health risk without wearable physiological monitoring. Perceived adaptive capacity showed consistent negative correlations with stress (BPM: r=−0.44), indicating that both environmental conditions and psychological factors (perceived control, risk assessment) modulate adaptation.

4.3. Temporal dynamics and spatial aggregation for design validation

Extended adaptation durations demonstrate that complex multi-domain environments require substantial physiological stabilization time, with direct implications for thermal refuge design where brief exposure may provide insufficient adaptive benefit. PM-low frequency sound combinations showed particularly high recovery capacity (EDA: 0.894), representing frequency-specific responses missed by single-domain assessment yet actionable for landscape interventions. The hierarchical adaptation mechanism, cardiovascular responses showing broad sensitivity while electrodermal activity exhibits stronger thermal-air quality coupling, confirms that climate-resilient design requires orchestrating thermal, acoustic, and air quality interventions as integrated systems. Spatial clustering identified five statistically distinct adaptation zones (ANOVA: $F_{5,158}$=9.42, $p < 0.001$) corresponding to measurable urban typologies. Optimal zones (resilience=0.66±0.17) exhibited balanced air quality (CO_2≈426 ppm), moderate acoustics [58], and thermal stability (UTCI≈30 °C), while stress-prone zones (0.41 ±0.11) showed elevated CO_2 and broadband acoustic loads. This demonstrates that aggregating multidimensional urban features enables neighborhood-scale comfort mapping identifying where physiological adaptation manifests versus where demographic models merely predict quality of life [59].

4.4. Limitations and future directions

This study presents three primary limitations. First, the small sample size (eight participants) establishes proof-of-concept but constrains generalizability. However, the observed physiological heterogeneity within this homogeneous group validates individual-level monitoring as necessary rather than reflecting inadequate statistical power. Unmeasured covariates (sleep quality, caffeine, stress levels, menstrual phase) may contribute to observed heterogeneity alongside inherent physiological differences. Future work should incorporate ecological momentary assessment to partition person-specific traits from contextual modulators. Second, the two-day experimental period limits assessment of temporal stability in resilience metrics. We cannot validate whether observed patterns persist across diurnal cycles or seasonal acclimatization. However, the short timeframe achieves the primary objective: demonstrating that multi-domain models outperform single-domain indices in predicting acute physiological responses. Day-stratified analyses suggested that physiological heterogeneity was statistically significant within each session independently (Section 3.5, Table S4), indicating the temporal limitation of this study applies to seasonal generalisability rather than to the internal validity of the core findings.

Despite these limitations, this study establishes that multi-domain monitoring predicts physiological stress better than thermal indices alone, a finding unlikely to reverse with extended observation, though magnitude and individual stability require longitudinal verification.

5. Conclusion

Do we all feel the city the same way? This study suggests we do not by establishing a methodological infrastructure for outdoor environmental resilience assessment by integrating individual physiological sensing with spatial analysis to validate whether building and landscape interventions successfully reduce stress for actual occupants [60]. Despite sample and temporal constraints detailed in Section 4.4, this investigation demonstrates technical feasibility and addresses three foundational questions:

1. **Multi-Domain Superiority (RQ1)**: Multi-domain models combining thermal, acoustic, and air quality parameters explained up to 43% of physiological variance, while heat stress index (UTCI) alone produced no predictive power. This demonstrates that UTCI functions as a moderator of other stressors rather than an independent driver, fundamentally challenging single-domain building performance frameworks.
2. **Individual Heterogeneity (RQ2)**: In this pilot sample, individuals with similar characteristics showed statistically significant differences in physiological stress responses, with effect sizes large enough to be detectable at this sample size. While $n = 8$ does not allow broad generalisation, the results demonstrate that meaningful physiological heterogeneity exists even within a homogeneous group, supporting the need for larger studies to validate whether personalised monitoring improves vulnerability assessment beyond population-averaged models.
3. **Spatial Aggregation (RQ3)**: K-means clustering identified five statistically distinct adaptation zones ($p < 0.001$) corresponding to measurable urban typologies, demonstrating that personalized wearable data can be aggregated into neighborhood-scale resilience maps. This enables evidence-based validation of where interventions improve physiological adaptation versus where demographic models merely predict vulnerability.

The three resilience metrics (Table 1) revealed extended physiological stabilization periods, with direct implications for thermal refuge design where brief exposure may provide insufficient adaptive benefit. The hierarchical adaptation mechanism, for example cardiovascular responses showing broad sensitivity while EDA exhibits stronger thermal-air quality coupling, confirms that climate-resilient building design requires coordinating multisensory interventions as integrated systems. Yet current practice lacks methods to validate whether these strategies improve resilience for actual inhabitants versus theoretical models.

This framework provides a methodological pathway for closing the validation loop between building intervention and individual response. The three-resilience metrics: Toughness, Recovery Capacity, and Adaptation Duration, translate individual measurements into actionable planning parameters that complement demographic indices and GIS modeling, enabling empirical assessment of where population-calibrated design assumptions fail and which landscape configurations produce measurable stress reduction for actual inhabitants. By integrating physiological sensing with spatial analysis, this proof-of-concept framework demonstrates how individual-level measurements can validate whether landscape interventions produce measurable stress reduction at neighborhood scale. This approach addresses a critical gap in current practice: while building science optimizes thermal performance and urban planning maps demographic vulnerability, neither validates whether implemented designs actually reduce stress for people inhabiting those spaces [61]. Human physiological responses provide the missing validation layer connecting building-scale performance to neighborhood-scale climate outcomes.

Future scaling of this methodology and the integration of validated physiological stress models with physics-informed digital twins that combine building energy simulation and urban microclimate modeling would enable pre-deployment validation: testing whether coordinated building innovations (material assemblages, transitional space configurations, cooperative energy management) aggregate to measurable pedestrian comfort improvements before construction begins. This requires scaling from proof-of-concept validation ($n = 8$) to population-level inference, from two-day acute response to seasonal acclimatization monitoring, and resolution of technical barriers regarding sensor network scalability, real-time data processing infrastructure, and privacy governance frameworks [62]. Applications of the validated proof-of-concept framework include green infrastructure placement strategies informed by measured recovery capacity, thermal refuge network design calibrated to adaptation duration requirements, and evidence-based evaluation of whether densification interventions maintain or improve physiological resilience. The fundamental question remains: will climate adaptation continue relying on predicted comfort from population-averaged models, or begin validating interventions through measured physiological outcomes of actual inhabitants experiencing urban space?

CRediT authorship contribution statement

Christos Grapas: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marcel Schweiker:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Anna Laura Pisello:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Ilaria Pigliautile:** Writing – review & editing, Supervision, Investigation. **Roberta Jacoby Cureau:** Writing – review & editing, Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.buildenv.2026.114679.

Data availability

Data will be made available on request.

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